Reflection

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1. Introduction

This reflection summarizes my learning process, challenges, and personal growth throughout the development of my Secure Notepad Web Application project.

The project's goal was to design and build a secure web-based note-taking app that applied encryption, authentication, and secure hosting principles — a practical exploration of cybersecurity fundamentals.

2. Learning Objectives

My main learning goals were to:

- Understand and apply **encryption and hashing** methods for data security.
- Learn how to **host and secure** a web application with HTTPS and SSL/TLS certificates.
- Improve my **web development and database management** skills using Flask and SQLite.

- Gain experience in **system deployment and server administration** using a Raspberry Pi.

These objectives guided my focus and provided measurable outcomes throughout the project.

3. Technical Growth

This project significantly expanded my technical foundation. I gained practical experience in:

- Setting up a **Flask web server** and managing its routing structure.
- Implementing **user authentication** with hashed credentials using Werkzeug security.
- Understanding the differences between **AES, RSA**, and other encryption algorithms, even if I implemented a simpler substitution cipher for proof of concept.
- Using **Certbot** to automate SSL/TLS certificate creation and renewal for HTTPS.
- Managing a **Raspberry Pi** as a web host - configuring Nginx, opening ports, and troubleshooting deployment issues.
- Applying **GitHub for version control**, learning how to document and track changes efficiently.

Each of these skills contributes directly to my broader goal of pursuing cybersecurity, combining both theoretical understanding and hands-on application.

4. Problem-Solving Adaptation

The project did not follow a straight path. Initially, I planned to create a **desktop application**, but after feedback from **Xuemei**, I pivoted to a **web-based solution** to gain more exposure to different technologies and security practices.
Later, **Ben's advice** about not storing encryption keys locally pushed me to rethink data flow and security architecture.

The most challenging parts were:

- Understanding encryption concepts deeply enough to implement them.
- Setting up the server and HTTPS for the first time.
- Managing time effectively while troubleshooting technical issues.

Despite these challenges, each obstacle taught me something crucial - from better researching encryption libraries to structuring my code for maintainability and scalability.

5. Collaboration and Feedback

Teacher feedback was instrumental to my progress:

- **Xuemei** guided the project's direction early on, steering it toward a web application and HTTPS implementation.
- **Ben** challenged my security assumptions, inspiring improvements in key management and data handling.
- **Ceryl** helped me with **Nginx setup** and resolving deployment issues on my Raspberry Pi.
- **Petra** gave useful post-presentation advice to simplify user interaction and refine the app's design.

Peer discussions also helped validate design decisions and improve usability.
Overall, collaboration shaped the project's evolution from a simple idea into a fully hosted web app.

6. Personal Development

Beyond technical growth, this project improved several personal skills:

- **Public Speaking:** Presenting my project to Petra and the class taught me how to stay composed under pressure and explain complex ideas clearly.
- **Time Management:** Balancing development, debugging, and research helped me structure my workflow better.
- **Resilience:** I learned how to stay patient through setbacks, especially when dealing with deployment issues.
- **Confidence:** Seeing the final product working live gave me confidence in my abilities as a developer and problem solver.

7. Conclusion

The Secure Notepad Web Application project was a defining learning experience.It combined software development, security, networking, and self-management into one cohesive challenge.

Despite technical difficulties and time constraints, the final result demonstrates my ability to apply cybersecurity concepts in practice, adapt to feedback, and deliver a functional product.

This experience not only solidified my interest in cybersecurity but also gave me the confidence and foundation to tackle more advanced projects in the future.